BTS– III – 11.13. 1186

B.Tech. Degree III Semester Examination November 2013

ME 1303 MECHANICS OF SOLIDS

(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

($8 \times 5 = 40$)

I. (a) Draw stress-strain diagram for ductile and brittle materials and indicate salient points.

(b) Explain the significance of Mohr's Circle and its uses.

(c) A circular shaft is subjected to a torque of 10kNm. The power transmitted by the shaft is 209.33KW. Find the speed of shaft in rpm.

(d) What do you mean by point of contra flexure? Is the point of contra flexure and point of inflexion different?

(e) What do you understand by neutral axis and moment of resistance? How do you locate neutral axis?

(f) Sketch and explain the shear stress distribution in a symmetrical I-Section.

(g) Obtain the maximum value of deflection for a cantilever of length 'L', constant 'EI' and carrying concentrated load 'W' and the end.

(h) Discuss the effect of the slenderness ratio of a column over buckling load.

PART B

($4 \times 15 = 60$)

II. A compound tube consists of steel tube 170mm external diameter and 10mm thickness and an outer brass tube 190mm external diameter and 10mm thickness. The two tubes are of equal length. The compound tube carries an axial load of 1MN. Find the stresses and the load carried by each tube and the amount by which it shortens. Length of each tube is 200mm. $E_{steel} = 200GN/m^2$ and $E_{Brass} = 100\ GN/m^2$.

OR

III. The normal stresses acting on two perpendicular planes at a point in a strained material are $70MN/m^2$ $(tensile)$ and $35\ MN/m^2$ $(compressive)$. In addition, shear stress of $40MN/m^2$ act on these planes. Calculate,

i) The magnitude of principal stresses
ii) The direction of principal stresses
iii) The magnitude of the maximum shear stress

IV. A solid circular shaft transmits 75 KW power at 200rpm. Calculate the shaft diameter, if the twist in the shaft is not to exceed one degree in 2m length of the shaft and shear stress is not to exceed $50N/mm^2$. Assume the modulus of rigidity of the material of the shaft as $100\ KN/mm^2$.

OR

(P.T.O.)

V. An overhanging beam ABC is simply supported at A and B over a span of 6m and BC overhangs by 3m. If the supported span AB carries central concentrated load of 8kN and over hanging span BC carries 2 kN/m completely. Draw shear force and bending moment diagrams indicating salient points.

VI. A simple beam of span 10m carries a UDL of 3kN/m. The section of the beam is a 'T' having a flange of 125x125mm and web 25x175mm. For the critical section obtain the shear stress at the neutral axis and at the junction of the flange and the web. Also draw the shear stress distribution across the section.

OR

VII. Discuss on various theories of failures and their applications to ductile and brittle materials.

VIII. A beam of length 10m is simply supported at its ends and carries two point loads of 100kN and 60kN at a distance of 2m and 5m respectively from the left support. Calculate the deflections under each load. Find also the maximum deflection also. Take $I = 18 \times 10^8 mm^4$ and $E = 2 \times 10^5 N/mm^2$.

OR

IX. A column of solid circular section 12cm diameter, 3.6m long is hinged at both ends. Rankine's constant is $^1/1600$ and $\sigma_c = 54kN/cm^2$. Find the buckling load. If another column of the same length, end conditions, and Rankine's constant but of 12cmx12cm square cross-section, and different material, has the same buckling load. Find the value of σ_c of its material.

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